

Sales Transaction Data Dictionary

Data Fields Reference

Transaction Identifiers

Field	Description	Data Type	Purpose
BranchName	Name/identifier of the branch or location where the sale occurred	Text	Geographic segmentation, location-based price analysis
OrderNumber	Unique identifier for the complete customer order	Text/Numeric	Groups multiple line items that were purchased together
InvoiceNumber	Unique identifier for the billing document	Text/Numeric	Links to financial records and payment processing
LineNumber	Sequential number identifying each product line within an order	Numeric	Distinguishes individual items within multi-item orders

Temporal Data

Field	Description	Data Type	Purpose
OrderDate	Date when the customer placed the order	Date	Seasonal analysis, demand patterns, time-based price sensitivity
SalesDate	Date when the sale was completed/delivered	Date	Fulfillment timing, actual revenue recognition date

Personnel Information

Field	Description	Data Type	Purpose
SalesPersonID	Unique identifier for the sales representative	Text/Numeric	Performance tracking, sales method analysis

SalesPersonName	Name of the sales representative	Text	Human-readable reference for sales attribution
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Product Information

Field	Description	Data Type	Purpose
ProductID	Unique identifier for the specific product/SKU	Text/Numeric	Individual product price sensitivity analysis
ProductGroupID	Identifier for product category or group	Text/Numeric	Category-level price analysis, cross-elasticity studies

Customer Information

Field	Description	Data Type	Purpose
CustomerID	Unique identifier for the customer	Text/Numeric	Customer-level price sensitivity, loyalty analysis
CustomerName	Name of the customer	Text	Human-readable customer reference

Financial Metrics

Field	Description	Data Type	Purpose
ExtPrice	Extended price (UnitPrice × SalesQty) - total revenue for this line item	Currency	Total revenue analysis, price point effectiveness
ExtCost	Extended cost (UnitCost × SalesQty) - total cost for this line item	Currency	Profitability analysis, margin impact of pricing
UnitPrice	Price charged per unit of the product	Currency	Core variable for price sensitivity analysis
UnitCost	Cost per unit of the product	Currency	Margin analysis, pricing floor determination

Quantity and Measurement

Field	Description	Data Type	Purpose
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SalesQty	Number of units sold	Numeric	Core variable - demand response to price changes
UOM	Unit of Measure (e.g., "each", "box", "case")	Text	Standardizes quantity comparisons across products
UnitSize	Size or volume of each unit	Numeric	Price per standardized unit calculations

Quotes Data Dictionary

Data Fields Reference

Quote Identifiers

Field	Description	Data Type	Purpose
QuoteID	Unique identifier for each quote request	Text/Numeric	Groups all quote versions for a single customer inquiry
QuoteVersion	Version number of the quote (1, 2, 3, etc.)	Numeric	Tracks price negotiation iterations - critical for analyzing price flexibility

Temporal Data

Field	Description	Data Type	Purpose
QuoteDate	Date when the quote was created/sent	Date	Timing analysis, seasonal quote patterns
ExpirationDate	Date when the quote expires	Date	Quote validity period, urgency effects on conversion
ConversionDate	Date when quote was converted to order (if won)	Date	Time-to-close analysis, decision timing patterns

Personnel & Location

Field	Description	Data Type	Purpose
SalesPersonID	Unique identifier for the sales representative	Text/Numeric	Sales performance, negotiation skill analysis
SalesPersonName	Name of the sales representative	Text	Human-readable sales attribution
BranchName	Branch/location where quote originated	Text	Geographic price variation analysis

Customer Information

Field	Description	Data Type	Purpose
CustomerID	Unique identifier for the customer	Text/Numeric	Customer-specific price sensitivity profiling
CustomerName	Name of the customer	Text	Human-readable customer reference

Product Information

Field	Description	Data Type	Purpose
ProductID	Unique identifier for the quoted product	Text/Numeric	Product-specific price acceptance analysis
ProductGroupID	Product category identifier	Text/Numeric	Category-level price sensitivity patterns

Quote Details

Field	Description	Data Type	Purpose
QuoteQty	Quantity of units quoted	Numeric	Demand quantity at quoted price point
UnitPrice	Price per unit quoted to customer	Currency	Core variable - proposed price point
UOM	Unit of Measure	Text	Standardizes quantity comparisons
UnitSize	Size/volume per unit	Numeric	Price per standardized unit calculations

Outcome Tracking

Field	Description	Data Type	Purpose
QuoteStatus	Final outcome of the quote	Text	Critical for price sensitivity analysis
ConvertedToOrderNumber	Order number if quote was won	Text/Numeric	Links to sales transaction data

Quote Status Values & Meanings

'Won' - Quote Accepted

- Customer accepted the quoted price
- Quote converted to a sales order
- **ConvertedToOrderNumber** populated
- **ConversionDate** shows when deal closed
- Represents **successful price points**

'Revised' - Price Negotiation Required

- Original quote price was **too high** for customer
- Sales team had to **reduce price** to secure deal
- Multiple versions exist (QuoteVersion > 1)
- Final version should show 'Won' status
- **Key indicator of price resistance**

'Lost' - Quote Rejected

- Customer rejected the final quoted price
 - No sale resulted from this quote
 - **ConvertedToOrderNumber** is null/empty
 - **Critical data points showing customer price limits**
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Quote Negotiation Patterns

None

QuoteID = "Q001"

└─ Version 1: \$100/unit → Status: 'Revised' (too high)

└─ Version 2: \$90/unit → Status: 'Revised' (still too high)

└─ Version 3: \$85/unit → Status: 'Won' (acceptable price)

Price Acceptance Thresholds

- **Immediate Wins** (QuoteVersion = 1, Status = 'Won'): Price below customer's threshold
- **Negotiated Wins** (QuoteVersion > 1, Status = 'Won'): Final acceptable price after revisions
- **Price Ceilings** (Status = 'Lost'): Customer's maximum acceptable price point

Inventory Data Dictionary

Data Fields Reference

Field	Description	Data Type	Purpose
BranchCode	Unique code identifier for the branch/location	Text/Numeric	Precise branch identification, links to other tables
BranchName	Descriptive name of the branch/location	Text	Human-readable location reference
ProductID	Unique identifier for the product/SKU	Text/Numeric	Links to sales and quotes data for supply-demand analysis
QtyOnHand	Current quantity of product available at this branch	Numeric	Stock availability - critical control variable

Critical Role in Price Sensitivity Analysis

1. Supply Constraint Identification

Problem Without Inventory Data:

None
Product A sales dropped 50% after price increase
Conclusion: High price sensitivity ❌

Analysis With Inventory Data:

None

Product A sales dropped 50% after price increase

BUT: QtyOnHand = 0 for 3 weeks during analysis period

Conclusion: Sales drop due to stock-out, not price sensitivity ✓

2. Demand vs. Availability

- **True Demand:** Sales limited by customer price sensitivity
 - **Constrained Demand:** Sales limited by inventory availability
 - **Unconstrained Analysis:** Only analyze periods/products with adequate stock
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Key Analysis Applications

Stock-Out Impact Analysis

Identify instances where demand might have been higher:

SQL

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-- Example logic for analysis
IF QtyOnHand = 0 AND QuoteQty > 0
THEN demand was constrained by supply, not price
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Inventory Turnover by Price Point

- **Fast turnover + high prices** = Strong demand despite premium pricing
- **Slow turnover + low prices** = Price-insensitive or weak demand
- **Zero turnover + adequate stock** = Price point too high

Branch-Level Supply Analysis

Compare sales performance across branches:

None

Branch A: High sales, Low inventory → Strong demand (price acceptable)

Branch B: Low sales, High inventory → Weak demand (price resistance?)

Branch C: Low sales, Zero inventory → Demand unknown (supply constrained)

Integration with Sales & Quotes Data

Combined Analysis Framework

1. Sales Data Integration

None

Match: BranchName + ProductID

Analyze: SalesQty vs QtyOnHand

Purpose: Identify supply-constrained vs demand-limited sales

2. Quotes Data Integration

None

Match: BranchName + ProductID

Analyze: Lost quotes where QtyOnHand > QuoteQty

Purpose: True price resistance (had stock but customer rejected price)